

Econometrics Stata Lab: Compare a School with Itself

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Week 4, Fall 2026

Motivating question: Papke (2005), "The effects of spending on test pass rates: evidence from Michigan."

We begin with pictures and school stories; regression terms come later.

Outline

1. **The question and the data:** what one row is; Papke's question.
2. **Three made-up schools:** one line through all dots vs. each school's own path.
3. **Fixed effects by hand:** subtract each school's average; then `xtreg, fe`.
4. **Dummy variables:** one intercept per school gives the same answer.
5. **Next week:** the same tools on 1,773 real Michigan schools.

What you should be able to explain

By the end of this lecture, you should be able to:

1. say what one row of the school data represents;
2. point to the comparison that follows one school over time;
3. explain what disappears when we subtract a school's usual level; and
4. explain how a dummy variable gives each school its own intercept.

The paper's question

Papke (2005) asks: **when a Michigan school gets more spending per pupil, do more fourth-graders pass the math test?**

Comparison	What we ask	Example
Between schools	Does a school that spends more score higher?	Maple (8.4) vs Pine (7.9) in 1994
Within a school	In years when a school spends more, does it score higher than it usually does?	Maple 1994 (8.4) vs Maple 1997 (8.7)

A small example before the regression

Three made-up schools, four years each. In every school, scores rise as that school's spending rises. Follow one color from year to year.

Maple (high spend, low scores)			
Year	\$ per pupil	lavgrexpp	math4
1994	4,450	8.4	42
1995	4,910	8.5	43
1996	5,430	8.6	44
1997	6,000	8.7	45

Pine (low spend, high scores)			
Year	\$ per pupil	lavgrexpp	math4
1994	4,450	8.4	42
1995	4,910	8.5	43
1996	5,430	8.6	44
1997	6,000	8.7	45

Oak sits in the middle (not shown). Start by tracing each school's own path.

Variable	Meaning	Role
math4	% of 4th graders with a satisfactory score on the state math test	outcome
lavgrexpp	log of real spending per pupil, averaged over this and last year	main variable

Log vs dollars, for Maple in 1994: $\ln(4,450) \approx 8.4$, and going back, $e^{8.4} \approx 4,450$. Each 0.1 step in the log is roughly a 10% rise in dollars.

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Year	\$ per pupil	lavgrexpp	math4	Year	\$ per pupil	lavgrexpp	math4
1994	4,450	8.4	42	1994	2,700	7.9	68
1995	4,910	8.5	43				
1996	5,430	8.6	44				
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Year 1: Maple spends more (8.4 vs 7.9) but scores lower (42 vs 68).

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Year 2: The same schools move a little; the large gap between them remains.

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1996	5,430	8.6	44	1996	3,290	8.1	70
1997	6,000	8.7	45				

Year 3: Within each school, spending and scores have moved up together.

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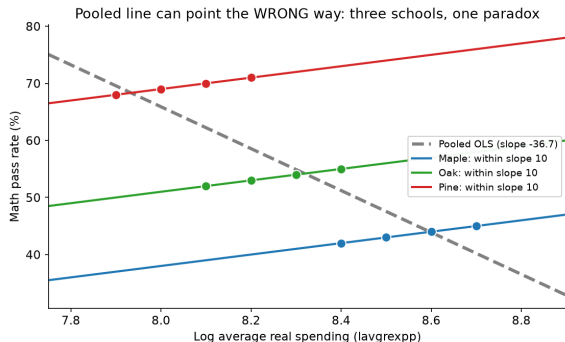
Pine (low spend, high scores)			
Year	\$ per pupil	lavgrexpp	math4
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1995	2,980	8.0	69
1996	3,290	8.1	70
1997	3,640	8.2	71

Year 4: The pattern within a school points up; comparing schools points down. Which comparison answers Papke's question?

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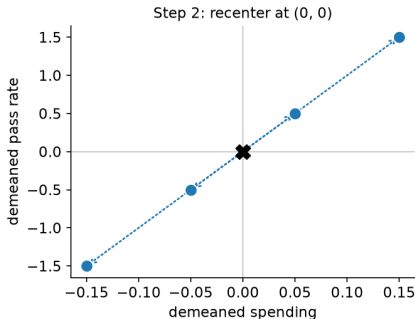
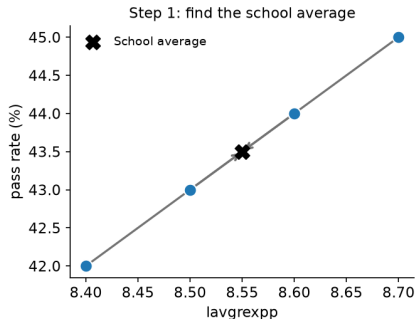
One line through every dot can tell the wrong story



The grey dashed line connects all school-year observations. In this example it slopes down. Each colored line follows one school, and every school's path slopes up. The all-dot line is influenced by the different usual levels of the three schools.

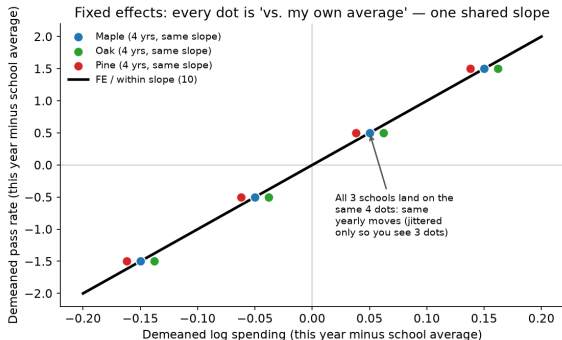
Demeaning in two steps (Maple shown)

Demeaning = subtracting the school's own center (Maple shown)



Step 1: find the school average (the $\times$). Step 2: every dot becomes “this year minus my average.” Later we will call this the **within transformation**. In Stata, `xtreg ... , fe` does this work for us.

Now compare each school with its own usual level



For each school, subtract its average spending and its average score from every year's values. Each school is now centered around (0,0). The comparison uses changes from that school's usual level, so the gap between schools disappears.

Now name the two comparisons

Pooled OLS

One line through all school-years. It combines differences across schools with changes inside each school.

School fixed effects

Subtract each school's usual level, then compare its year-to-year changes. A lasting difference between schools drops out. **BUT** *a change in student mix, school leadership, or another policy does not automatically drop out.*

Dummy variables: 1 if true, 0 if not

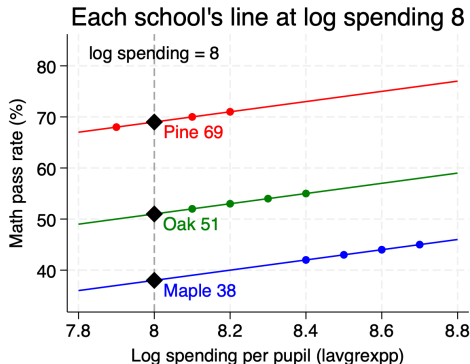
A **dummy** (indicator, binary) variable is 1 when something is true and 0 otherwise.

school	year	maple	oak	pine
Maple	1994	1	0	0
Maple	1995	1	0	0
Oak	1994	0	1	0
Oak	1995	0	1	0
Pine	1994	0	0	1
Pine	1995	0	0	1

Every row has exactly one 1: each school-year belongs to one school.

In Stata: `generate oak = (school == "Oak")`. The statement in brackets is 1 when true and 0 when false.

Where does each school's line sit at log spending 8?



Extend each school's line (slope 10) to log spending 8, a level inside our data (about \$3,000 per pupil).

Where the lines cross the dashed line:

- Maple 38
- Oak 51
- Pine 69

These heights are what the dummies will estimate. To make them the regression's intercepts, move zero to 8:

$$\text{spend8} = \text{lavgrexpp} - 8$$

First: run each school on its own

Regress math4 on spend8 using only one school's four rows:

Regression	Intercept	Slope
<code>regress math4 spend8 if school == "Maple"</code>	38	10
<code>regress math4 spend8 if school == "Oak"</code>	51	10
<code>regress math4 spend8 if school == "Pine"</code>	69	10

Three regressions, three intercepts (the heights at log spending 8), one slope: 10. Next: get all three intercepts from **one** regression, using dummies.

Separate regressions also let each school have its own *slope*; the dummy regression will force one shared slope. They agree here because the toy data were built with slope 10 in every school. With real data, fixed effects combines the schools into one average within-school slope.

Dummies move the intercept, not the slope

With $\text{spend8} = \text{lavgrexpp} - 8$, regress with two of the three dummies (Maple left out):

$$\widehat{\text{math4}} = 38 + 13 \text{ oak} + 31 \text{ pine} + 10 \text{ spend8}$$

Switch the dummies on and off, one school at a time:

School	oak	pine	Intercept	Slope
Maple	0	0	38	10
Oak	1	0	$38 + 13 = 51$	10
Pine	0	1	$38 + 31 = 69$	10

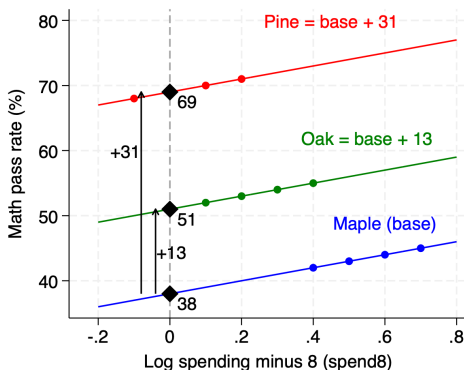
- **Same slope** (10) in every line: the effect of spending.
- **Different intercepts** (38, 51, 69: pass rate at $\text{spend8} = 0$): each school's usual level, its **fixed effect**. A dummy shifts the whole line up or down.

The dummy variable trap

Why leave Maple out?

- On every row, $\text{maple} + \text{oak} + \text{pine} = 1$, just like the constant.
- So `maple` adds nothing new: not Oak and not Pine means Maple. Stata drops one.
- **Rule:** k groups $\Rightarrow k - 1$ dummies.

The school left out is the **base**: its line is the constant (38). Each dummy says how far that school's line sits *above* the base: Oak +13, Pine +31.



Same slope (10); only the height of the line moves.
Diamonds = intercepts at $\text{spend8} = 0$.

Let Stata make the dummies: the `i.` prefix

`regress math4 spend8 i.school_id` makes one dummy per school and leaves the first (Maple) out as the base. Same output: constant 38, Oak +13, Pine +31, slope 10.

Way of estimating	Slope
Pooled OLS (compare different schools)	-36.7
Subtract each school's average, then OLS	10
<code>xtreg math4 lavgrexpp, fe</code>	10
Dummies by hand (oak, pine)	10
<code>i.school_id</code>	10

Every way of comparing a school with *itself* gives 10.

How to read the slope of 10

`xtreg math4 lavgrexpp, fe` gives a slope of **10**. It compares each school with **itself**, never Maple with Pine.

In words

When a school's spending rises by about 10% (0.1 in logs), that same school's pass rate rises by about $10 \times 0.1 = 1$ percentage point.

Check it on Maple: 1994 $\rightarrow$ 1995, spending \$4,450 $\rightarrow$ \$4,910 (`lavgrexpp` 8.4 $\rightarrow$ 8.5), and `math4` 42 $\rightarrow$ 43.

Why not “+1 in the log”? Adding 1 to the log multiplies dollars by $e \approx 2.7$ (about +172%), which no school does. Adding 0.1 multiplies dollars by $e^{0.1} \approx 1.105$: about 10%.

The rest of the `xtreg` output can wait until next week.

Run it in Stata

The Week 4 do-file `week04_pooled_vs_fe.do` runs, top to bottom:

1. Pooled OLS: `regress math4 lavgrexpp`
2. Demeaning by hand: `bysort school: egen, generate, regress`
3. Fixed effects: `xtset school_id year, xtreg math4 lavgrexpp, fe`
4. Dummies: `generate oak = (school == "Oak"), generate spend8 = lavgrexpp - 8, regress math4 spend8 oak pine, then i.school_id`

Change the `cd` line to your own week-04 folder first. The log and the graphs are saved there.

Next week: real Michigan schools

Papke (2005): 1,773 Michigan schools, 1994–1998, after the 1994 funding reform.

- The same two comparisons: pooled OLS vs. school fixed effects.
- **Year dummies** (i.e. *year*): exactly like today's school dummies, but one intercept per *year*, to remove changes that hit every school at once.
- Reading a regression table: standard errors, stars, confidence intervals.

Bring today's idea: **compare each school with itself**.

Tell me how the course is going

- Anonymous, about 1 minute, all questions optional.
- Pick the week, your section, and rate the pace.
- Fill it in once for each week, if you like.

[Open the feedback form](#)



Scan with your phone camera

Appendix: fixed effects = OLS on demeaned data

Each school i has its own usual level α_i (no t : it never changes):

$$\text{math4}_{it} = \alpha_i + \beta \text{lavgrexpp}_{it} + u_{it}$$

Average each school over its years:

$$\overline{\text{math4}}_i = \alpha_i + \beta \overline{\text{lavgrexpp}}_i + \bar{u}_i$$

Subtract the second line from the first. $\alpha_i - \alpha_i = 0$:

$$\underbrace{\text{math4}_{it} - \overline{\text{math4}}_i}_{\text{this year minus my average}} = \beta \underbrace{(\text{lavgrexpp}_{it} - \overline{\text{lavgrexpp}}_i)}_{\text{this year minus my average}} + (u_{it} - \bar{u}_i)$$

The usual level is gone. OLS on the demeaned data estimates β : the **within transformation**.

Appendix: the same steps in Stata

Step	Stata (Maple, Oak, Pine)
School average $\bar{x}_i$	<code>bysort school: egen mean_spend = mean(lavgrexpp)</code>
This year minus average	<code>generate dm_spend = lavgrexpp - mean_spend</code>
Same for the outcome	<code>generate dm_math4 = math4 - mean_math4</code>
OLS on demeaned data	<code>regress dm_math4 dm_spend → slope 10</code>
All of it in one command	<code>xtreg math4 lavgrexpp, fe → slope 10</code>

Same slope. `xtreg`'s standard errors are a little bigger: once you know a school's average and 3 of its 4 years, the 4th year adds nothing new, so each school has only 3 free observations ($12 - 3 - 1 = 8$ degrees of freedom, not 10).